

# A new group contribution model for prediction liquid hydrocarbon viscosity based on free-volume theory

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## ARTICLE INFO

### Article history:

Received 5 September 2022

Revised 7 February 2023

Accepted 9 February 2023

Available online 13 February 2023

### Keywords:

Group contribution model

Viscosity

n-alkanes

Free-Volume Theory

## ABSTRACT

In this paper a group contribution model for predicting the viscosity values of n-alkanes and iso-alkanes in the liquid phase based on Free-Volume Theory was developed. A database with 2129 experimental data with a temperature range up to 573.15 K and a pressure range up to 4000 bar was used to develop the model. The total weighted average percentage error is 4.894%. The model was tested on the next 472 experimental data with a temperature range up to 564.15 K and a pressure range up to 2019 bar and a mean percentage error that was obtained is 5.155%.

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## 1. Introduction

Values of fluid transport properties are necessary for the design of many industrial processes, such as the design of heat exchangers, distillation columns and other apparatus in the chemical, petrochemical and food industries, as well as for the design of pipelines. Knowing the value of viscosity, both pure substances and mixtures, is of special importance for the mentioned processes. The viscosity of the substance indicates the possibility of handling, storage and disposal. Therefore, viscosity has been extensively investigated experimentally for various groups of fluids, such as hydrocarbons, alcohols, esters, ethers, acids, amines, ionic liquids and other groups of compounds over a wider range of temperatures and pressures, as well as compositions for mixtures.

However, on the one hand, experimental measurements are expensive and time consuming, and on the other hand, when designing processes, it is often necessary to know the value of viscosity at intermediate values, especially temperature and pressure, i.e. the composition of the mixture. Therefore, many models have been developed, both for correlating [1–13] and for predicting [14–73] viscosity values, which enable the calculation of this values at the required temperature, pressure and composition. Some of these models have universal applicability, while the applicability of others is limited to one or more classes of substances.

Experimental measurements are required for the development of correlation models, and the calculation of viscosity values is limited to the temperature range as well as to the pressure range. Prediction models require knowledge of the characteristic values of the substance, such as critical quantities and acentricity factors, most often. A special group of prediction models are models of group contributions, which are based on knowledge of the values of the groups that make up the substance. To calculate the value of groups, you need as extensive database as possible.

The aim of this paper is to develop a group contribution model for predicting the viscosity values of liquid hydrocarbons based on Free-Volume theory.

## 2. New model

Most prediction models have been developed for specific values of temperature and pressure, i.e. for room temperature and atmospheric pressure, or for specific groups of compounds.

All correlation models, except the Lee-Wei [7] model, are non-linear in parameters, but models Ducoulombier-Zhou-Boned-Peyre lasse-Saint-Guirons-Xans I [9], Ducoulombier-Zhou-Boned-Peyre lasse-Saint-Guirons-Xans II [9] and Comunas-Baylaucq-Boned-Fernandez [10] can be linearized. Allal-Boned-Baylaucq [13] model can also be applied to gases. In addition to the values of the densities in models Assael-Dalaouti-Polimatidou [12], Assael-Dalaouti [11] and Assael-Dalaouti-Dymond [5], it is necessary to know the values of the critical temperature and the critical density. In the Kiran-Sen [8] model, knowledge of the density values is required, and in the Lee-Wei [7] model, the critical temperature.

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The Souders model [14], Kreps-Druin [18], Orbey-Sandler [28], and Conte-Martinho-Matos-Gani [57] are applicable only at atmospheric pressure, the Pachaiyappan-Ibrahim-Kuloor model [17] at temperature  $T = 293.15$  K and atmospheric pressure, the Dymond model [19] at low pressure up to a pressure of 1 bar, the Tatev-Abramenskoy-Grikina model [28] to alkanes, alkenes, alkylcyclopentanes, alkylcyclohexanes and alkylbenzenes at atmospheric pressure, the Kanti-Zhou-Ye-Boned-Lagourette-Saint-Guiron-San s-Montel model [30] to alkanes and alkylbenzenes and with knowledge of dynamic viscosities values at reference temperature and pressure of 1.01325 bar, the Marano-Holder model [31] to n-alkanes and n-alkenes at atmospheric pressure, the Gürbüz Yücel-Özdoğan model [69] at atmospheric pressure, the Riazi-Enezi-Soleimani model [37] to n-alkanes at atmospheric pressure and with knowledge of refractive indices values, the Marrero-Morejón-Pardillo-Fontdevila model [35] at  $T = 293.15$  K and atmospheric pressure, the Korsten model [56] to two temperatures (313.15 and 373.15 K) and atmospheric pressure, the

Hsu-Sheu-Tu model [34] at atmospheric pressure and for temperatures for which  $T_r < 0.75$ , the Riazi-Mahdi-Alqallaf model [41] to hydrocarbons ( $11 \leq n_c \leq 20$ ) at atmospheric pressure and with knowledge of refractive indices values. The Ceriani-Conçalves-Rabelo-Caruso-Cunha-Cavaleri-Batista-Meirelles [55] and Ceriani-Conçalves-Coutinho [54] models are intended for predicting the values of dynamic viscosity of esters, acylglycerols, fatty acids and alcohols with large molecules and oligomeric molecules, respectively. The Matsuda-Yamamoto-Kurihara-Tochigi model [53], the Gardas-Coutinho I [42], Gardas-Coutinho II [42] and Gardas-Ge-Goodrich-Hardacre-Hussain-Rooney [44] models are evaluated for ionic liquids. The Cano-Gómez-Iglesias-Silva-Ramos-Estrada-Hall I [49] and Cano-Gómez-Iglesias-Silva-Ramos-Estrada-Hall II [49] models are evaluated for 1-alcohols at atmospheric pressure and with knowledge of refractive indices values and the Ciotta-Trusler-Riesco-Vesovic model [50] for n-alkanes. The Wang-Zhang-Li-Chen-Dong-Zhao-Zhong-Sheng-Li-Gong model [72] has been proposed for natural gas components based on the

**Table 1**

Sources of experimental data used in development a new model, number of compounds,  $l$ , minimum and maximum temperature values,  $T_{\min}$  and  $T_{\max}$ , minimum and maximum pressure values,  $p_{\min}$  and  $p_{\max}$ , and number of experimental data,  $N$ .

| $l$ | Compounds               | $T_{\min}/K$ | $T_{\max}/K$ | $p_{\min}/bar$ | $p_{\max}/bar$ | $N$ | References                                                              |
|-----|-------------------------|--------------|--------------|----------------|----------------|-----|-------------------------------------------------------------------------|
| 1.  | Propane                 | 173.15       | 273.15       | 68.950         | 344.740        | 30  | [74]                                                                    |
| 2.  | 2-Methylpropane         | 120.00       | 377.59       | 4.145          | 551.581        | 151 | [75,76]                                                                 |
| 3.  | Butane                  | 140.00       | 443.00       | 1.000          | 691.700        | 189 | [8,76,77]                                                               |
| 4.  | 2,2-Dimethylpropane     | 310.93       | 444.26       | 6.895          | 551.581        | 46  | [78]                                                                    |
| 5.  | 2-Methylbutane          | 273.13       | 293.15       | 1.013          | 1.013          | 2   | [79]                                                                    |
| 6.  | Pentane                 | 263.05       | 444.26       | 1.000          | 4000.000       | 219 | [8,79–88]                                                               |
| 7.  | 2,3-Dimethylbutane      | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 8.  | 2-Methylpentane         | 273.15       | 313.15       | 1.013          | 1.013          | 5   | [79,89,90]                                                              |
| 9.  | 3-Methylpentane         | 273.15       | 313.15       | 1.013          | 1.013          | 6   | [79,89–90]                                                              |
| 10. | Hexane                  | 273.15       | 448.00       | 1.013          | 4000.000       | 241 | [8,77,79,80,83–85,87,90–118]                                            |
| 11. | 2,2,3-Trimethylbutane   | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 12. | 2,2-Dimethylpentane     | 298.15       | 298.15       | 1.013          | 1.013          | 1   | [90]                                                                    |
| 13. | 2,3-Dimethylpentane     | 298.15       | 298.15       | 1.013          | 1.013          | 2   | [89,90]                                                                 |
| 14. | 2-Methylhexane          | 298.15       | 298.15       | 1.013          | 1.013          | 1   | [90]                                                                    |
| 15. | Heptane                 | 263.05       | 373.15       | 1.000          | 2488.800       | 133 | [79,82,84,85,87,90,91,97–99,105,107,108,117,119–129]                    |
| 16. | 2,2,3-Trimethylpentane  | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 17. | 2,2,4-Trimethylpentane  | 197.93       | 373.13       | 1.000          | 3119.000       | 152 | [79,89,90,96,97,108,130–139]                                            |
| 18. | 2,2-Dimethylhexane      | 273.15       | 313.15       | 1.013          | 1.013          | 4   | [79,90]                                                                 |
| 19. | 2,3,3-Trimethylpentane  | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 20. | 2,3,4-Trimethylpentane  | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 21. | 2,3-Dimethylhexane      | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 22. | 2,4-Dimethylhexane      | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 23. | 2,5-Dimethylhexane      | 273.15       | 313.15       | 1.013          | 1.013          | 4   | [79,90]                                                                 |
| 24. | 2-Methyl-3-ethylpentane | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 25. | 2-Methylheptane         | 273.15       | 273.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 26. | 3,3-Dimethylhexane      | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 27. | 3,4-Dimethylhexane      | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 28. | 3-Ethylhexane           | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 29. | 3-Methyl-3-ethylpentane | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 30. | 3-Methylheptane         | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 31. | 4-Methylheptane         | 273.15       | 313.15       | 1.013          | 1.013          | 3   | [79]                                                                    |
| 32. | Octane                  | 263.15       | 448.00       | 1.000          | 4000.000       | 325 | [8–79,80,82,85,87,90,97,98,100,104,105,107,108,117,121,123,127,140–148] |
| 33. | Nonane                  | 263.05       | 423.15       | 1.013          | 1.013          | 27  | [82,87,90,97,98,105,107,108,126,144]                                    |
| 36. | Decane                  | 283.15       | 444.26       | 1.000          | 2543.700       | 137 | [83,87,90,94,97,98,100,104–108,111,121,144,145,146,149,150]             |
| 35. | Undecane                | 263.05       | 473.15       | 1.013          | 1.013          | 7   | [82,151]                                                                |
| 36. | Dodecane                | 283.15       | 373.15       | 1.000          | 3670.000       | 71  | [87,90,97,105,107,108,140,142,145,146]                                  |
| 37. | Tridecane               | 267.67       | 473.15       | 1.000          | 1000.000       | 49  | [82,152]                                                                |
| 38. | Tetradecane             | 293.15       | 308.15       | 1.013          | 1.013          | 6   | [90,94,108,121]                                                         |
| 39. | Cyprane                 | 293.15       | 363.15       | 1.000          | 1000.000       | 142 | [134,153–156]                                                           |
| 40. | Hexadecane              | 293.15       | 373.24       | 1.000          | 3301.000       | 56  | [87,90,92,106,108,120,121,134,142]                                      |
| 41. | Heptadecane             | 295.15       | 573.15       | 1.013          | 1.013          | 8   | [82]                                                                    |
| 42. | Pristane                | 298.15       | 363.15       | 1.000          | 400.000        | 16  | [134,154]                                                               |
| 43. | Nonadecane              | 308.15       | 343.15       | 1.013          | 1.013          | 8   | [157]                                                                   |
| 44. | Eicosane                | 313.20       | 343.10       | 1.013          | 1.013          | 4   | [120]                                                                   |
| 45. | Docosane                | 323.10       | 343.20       | 1.013          | 1.013          | 3   | [120]                                                                   |
| 46. | Tetracosane             | 333.10       | 343.20       | 1.013          | 1.013          | 2   | [120]                                                                   |
| 47. | Octacosane              | 335.15       | 573.15       | 1.013          | 1.013          | 6   | [82]                                                                    |
| 48. | Squalane                | 273.15       | 333.15       | 1.000          | 300.000        | 18  | [106,134,143]                                                           |
| 49. | Dotriacontane           | 354.11       | 466.30       | 3.000          | 3.000          | 8   | [158]                                                                   |
| 50. | Hexatriacontane         | 373.15       | 573.15       | 1.013          | 1.013          | 5   | [82]                                                                    |

Free-volume Theory. The Polishuk-Yitzhak model [73] was developed for n-alkanes only. The Shukla [71], Goel [39] and Viswanath-Ghosh-Prasad-Dutt-Rani [40] models require knowledge of refractive indices values.

The Jossi-Stiel-Thodos [15,16], Jossi-Stiel-Thodos-Pedersen-Fredenslund-Christensen-Thomassen [24], Brulé-Starling-Chung-Ajlan-Lee-Starling [20,21,22], Batschinski-Hildebrand-Przedziecki-Sridhar [23], Tham-Gubbins-Pedersen-Fredenslund [25,26], Chung-Ajlan-Lee-Starling [20,21], Lee-Thodos [64], PR-Guo-Sun-Rong-Chen-Guo [36], Al-Syabi-Danesh-Tohidi-Todd-Tehrani [38], RKS-Quinones-Cisneros-Zéberg-Mikkelsen-Stenby [52], PR-Quinones-Cisneros-Zéberg-Mikkelsen-Stenby [52] and Kim-Lee [43] models require knowledge of critical values or normal boiling temperatures or normal freezing temperatures or acentric factors of the substances.

The Sun-Bleazard-Teja-Johnson-DiGiulio [61,67] and Bleazard-Teja-Sun-Smith-King [62,63,68] models are not intended to predict the i-alkane dynamic viscosity values.

### 3. Computational method

In this paper, a group contribution model for predicting liquid viscosity values, both n-alkanes and i-alkanes, was developed based on Free-Volume Theory [2].

$$\eta = \frac{\rho L(\alpha\rho + pM/\rho)}{\sqrt{3RTM}} \exp \left[ B \left( \frac{\alpha\rho + pM/\rho}{RT} \right)^{3/2} \right] \quad (1)$$

**Table 2**

The molecular groups contributions,  $\Delta L$ ,  $\Delta\alpha$  and  $\Delta B$ , and constant, A.

|                  | $\Delta L$ | $\Delta\alpha$ | $\Delta B \cdot 10^5$ |
|------------------|------------|----------------|-----------------------|
| A                | 0.9227     | 29.407         | 658.454               |
| -CH <sub>3</sub> | -0.02229   | 12.532         | -2.77587              |
| >CH <sub>2</sub> | 0.01217    | 9.8394         | -12.2326              |
| >CH-             | -0.1357    | 17.992         | -2.95564              |
| >C<              | -0.01725   | 4.2352         | 1.96344               |

**Table 3**

The absolute average relative deviation values per sets,  $p_{av}$ , and overall average relative deviation values,  $p_{pond}$  using proposed model for prediction of liquid hydrocarbon viscosity.

| <i>l</i>                                                       | Compounds               | $p_{av}/\%$ | <i>l</i> | Compounds               | $p_{av}/\%$ |
|----------------------------------------------------------------|-------------------------|-------------|----------|-------------------------|-------------|
| 1.                                                             | Propane                 | 6.474       | 26.      | 3,3-Dimethylhexane      | 7.837       |
| 2.                                                             | 2-Methylpropane         | 2.766       | 27.      | 3,4-Dimethylhexane      | 36.145      |
| 3.                                                             | Butane                  | 2.154       | 28.      | 3-Ethylhexane           | 38.315      |
| 4.                                                             | 2,2-Dimethylpropane     | 24.980      | 29.      | 3-Methyl-3-ethylpentane | 1.968       |
| 5.                                                             | 2-Methylbutane          | 9.485       | 30.      | 3-Methylheptane         | 21.409      |
| 6.                                                             | Pentane                 | 4.219       | 31.      | 4-Methylheptane         | 28.382      |
| 7.                                                             | 2,3-Dimethylbutane      | 3.862       | 32.      | Octane                  | 3.018       |
| 8.                                                             | 2-Methylpentane         | 16.558      | 33.      | Nonane                  | 2.022       |
| 9.                                                             | 3-Methylpentane         | 17.058      | 34.      | Decane                  | 3.232       |
| 10.                                                            | Hexane                  | 3.309       | 35.      | Undecane                | 5.930       |
| 11.                                                            | 2,2,3-Trimethylbutane   | 26.726      | 36.      | Dodecane                | 6.607       |
| 12.                                                            | 2,2-Dimethylpentane     | 0.986       | 37.      | Tridecane               | 5.170       |
| 13.                                                            | 2,3-Dimethylpentane     | 19.699      | 38.      | Tetradecane             | 8.997       |
| 14.                                                            | 2-Methylhexane          | 16.938      | 39.      | Cyprane                 | 5.173       |
| 15.                                                            | Heptane                 | 2.331       | 40.      | Hexadecane              | 10.821      |
| 16.                                                            | 2,2,3-Trimethylpentane  | 1.342       | 41.      | Heptadecane             | 7.343       |
| 17.                                                            | 2,2,4-Trimethylpentane  | 3.985       | 42.      | Pristane                | 15.379      |
| 18.                                                            | 2,2-Dimethylhexane      | 5.221       | 43.      | Nonadecane              | 5.431       |
| 19.                                                            | 2,3,3-Trimethylpentane  | 7.372       | 44.      | Eicosane                | 4.596       |
| 20.                                                            | 2,3,4-Trimethylpentane  | 16.575      | 45.      | Octacosane              | 4.803       |
| 21.                                                            | 2,3-Dimethylhexane      | 30.556      | 46.      | Docosane                | 1.345       |
| 22.                                                            | 2,4-Dimethylhexane      | 30.225      | 47.      | Tetracosane             | 9.359       |
| 23.                                                            | 2,5-Dimethylhexane      | 19.104      | 48.      | Squalane                | 10.540      |
| 24.                                                            | 2-Methyl-3-ethylpentane | 37.985      | 49.      | Dotriacontane           | 5.202       |
| 25.                                                            | 2-Methylheptane         | 14.555      | 50.      | Hexatriacontane         | 7.439       |
| Overall average relative deviation values $p_{pond}$ is 4.894% |                         |             |          |                         |             |

where  $\eta$  is the dynamic viscosity in  $\mu\text{P}$ ,  $M$  is the molar mass in  $\text{kg}/\text{mol}$ ,  $\rho$  is the density in  $\text{kg}/\text{m}^3$ ,  $p$  is the pressure in  $\text{MPa}$ , and  $R$  is the universal gas constant in  $\text{J}/(\text{mol}\cdot\text{K})$ , and  $L$  [m],  $\alpha$  [ $\text{J}\cdot\text{m}^3\cdot\text{mol}\cdot\text{kg}$ ],  $B$  are the parameters.

The values of the parameters in model (1) are calculated using the following relations.

$$L = A + \sum_{i=1}^4 n_i \Delta L_i \quad (2)$$

$$\alpha = A + \sum_{i=1}^4 n_i \Delta \alpha_i \quad (3)$$

$$B = A + \sum_{i=1}^4 n_i \Delta B_i \quad (4)$$

where  $A$  is a constant,  $n_i$  is the number of contributions of groups of the same type, and  $\Delta L$ ,  $\Delta\alpha$  and  $\Delta B$  are values of group contribution.

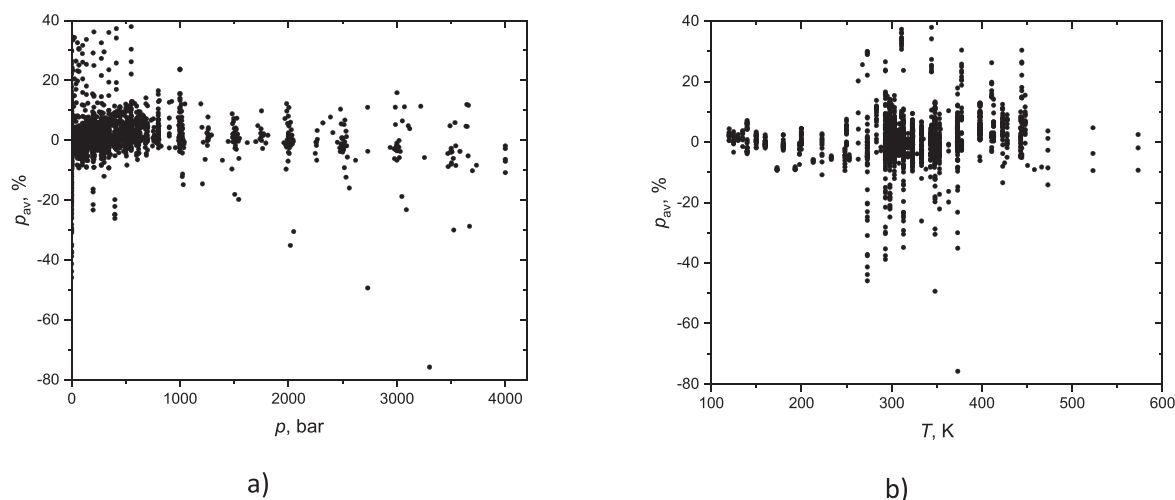
### 4. Results and discussion

To calculate the values of the group contributions in relations (2–4), we used experimental data for liquid viscosity, which are shown in Table 1. This table shows references of available data on liquid viscosity in the literature, as well as their temperature and pressure ranges, as well as number of experimental data.

The group contribution values for  $\Delta L$ ,  $\Delta\alpha$  and  $\Delta B$  and for constant  $A$  were obtained by applying the evolutionary algorithm, minimizing the overall average relative deviation. The calculated values of group contributions for all three parameters based on the experimental data from Table 1 are shown in Table 2.

The prediction results based on Eqs. (1)–(4) are shown in Table 3 and Fig. 1. The absolute average relative deviation values per sets  $p_{av}$  were calculated using Eq. (5).

$$p_{av}[\%] = (100/N) \sum_{i=1}^N \left| \left[ \eta_{i,\text{exp}} - \eta_{i,\text{cal}} \right] / \eta_{i,\text{exp}} \right| \quad (5)$$



**Fig. 1.** The absolute average relative deviation values per sets for prediction of liquid hydrocarbon viscosity in function of a) pressure and b) temperature.

where  $N$  is number of experimental points per set,  $\eta_{i,\text{exp}}$  experimental viscosity data, and  $\eta_{i,\text{cal}}$  calculated viscosity data using proposed prediction model. Overall average relative deviation values  $p_{\text{pond}}$  were obtained by applying the evolutionary algorithm for minimizing the overall average relative deviation Eq. (6).

$$p_{\text{pond}}[\%] = \frac{\sum_{i=1}^m N_i p_{av,i}}{\sum_{i=1}^l N_i} \quad (6)$$

where  $m$  is number of sets,  $N_i$  is number of experimental data and  $l$  is number of compounds.

Substances from Table 3 for which a deviation greater than 20% was obtained, with the exception of 2, 2 - dimethylpropane, are

represented with a small number of data  $n \sim 3$ , so this has a relatively small impact on the final result of 4.894%. For example: 2-methyl-3-ethylpentane has 3 experimental data points with  $p_{av}$  45.906, 37.561 and 30.487%, and overall  $p_{av}$  37.985%; 3-ethylhexane has also 3 experimental data points with  $p_{av}$  41.310, 38.784 and 34.852% and overall  $p_{av}$  38.315%. Allthrough overall  $p_{av}$  of hexadecane is 10.787%, maximum error are for  $T = 373.24$  K and  $p = 3301$  bar (75.782%) and for  $T = 348.15$  K and  $p = 2731$  bar (49.356%). No generally conclusion can be made, considering temperature range, pressure range and structure of alkanes, but best results are observed for n-alkanes.

For the substances in Table 1 with a sufficient number of experimental data (2073), a correlation was performed to compare with

**Table 4**

Correlation results, the molecular groups contributions,  $\Delta L$ ,  $\Delta \alpha$  and  $\Delta B$ , the absolute average relative deviation values per sets,  $p_{av}$ , and overall average relative deviation values,  $p_{\text{pond}}$ .

| $l$ | Compounds              | $L \cdot 10^{10}/\text{m}$ | $\alpha/[\text{J} \cdot \text{m}^3 \text{mol}^{-1} \cdot \text{kg}^{-1}]$ | $B \cdot 10^2$ | $p_{av}/\%$ |
|-----|------------------------|----------------------------|---------------------------------------------------------------------------|----------------|-------------|
| 1.  | Propane                | 1.050                      | 53.24708                                                                  | 0.7883         | 0.448       |
| 2.  | 2-Methylpropane        | 1.140                      | 54.79569                                                                  | 1.2129         | 2.061       |
| 3.  | Butane                 | 1.030                      | 63.65712                                                                  | 0.7946         | 1.983       |
| 4.  | 2,2-Dimethylpropane    | 0.738                      | 92.48559                                                                  | 0.9426         | 4.174       |
| 5.  | Pentane                | 1.430                      | 60.59355                                                                  | 0.7191         | 2.471       |
| 6.  | 2-Methylpentane        | 2.030                      | 39.52396                                                                  | 2.1456         | 0.555       |
| 7.  | 3-Methylpentane        | 1.940                      | 38.93023                                                                  | 2.3610         | 0.825       |
| 8.  | Hexane                 | 1.170                      | 80.35252                                                                  | 0.6405         | 2.141       |
| 9.  | Heptane                | 1.000                      | 95.29215                                                                  | 0.6617         | 1.315       |
| 10. | 2,2,4-Trimethylpentane | 0.800                      | 112.38250                                                                 | 0.6920         | 2.694       |
| 11. | 2,2-Dimethylhexane     | 0.429                      | 201.80609                                                                 | 0.2961         | 0.355       |
| 12. | 2,5-Dimethylhexane     | 2.030                      | 45.38770                                                                  | 2.4784         | 0.348       |
| 13. | Octane                 | 0.938                      | 114.41050                                                                 | 0.5774         | 3.017       |
| 14. | Nonane                 | 1.330                      | 86.59753                                                                  | 0.9924         | 1.917       |
| 15. | Decane                 | 0.829                      | 141.09900                                                                 | 0.5622         | 1.783       |
| 16. | Undecane               | 1.030                      | 124.81350                                                                 | 0.7449         | 2.916       |
| 17. | Dodecane               | 0.604                      | 199.64510                                                                 | 0.4251         | 3.141       |
| 18. | Tridecane              | 0.662                      | 207.51080                                                                 | 0.4175         | 2.105       |
| 19. | Tetradecane            | 0.453                      | 156.79057                                                                 | 0.8865         | 1.619       |
| 20. | Cyprane                | 0.644                      | 170.50912                                                                 | 0.7688         | 2.213       |
| 21. | Hexadecane             | 0.429                      | 291.44062                                                                 | 0.3292         | 1.921       |
| 22. | Heptadecane            | 0.595                      | 289.36200                                                                 | 0.3152         | 4.536       |
| 23. | Pristane               | 0.291                      | 318.65890                                                                 | 0.3815         | 6.998       |
| 24. | Nonadecane             | 0.313                      | 324.42350                                                                 | 0.3468         | 0.123       |
| 25. | Eicosane               | 1.820                      | 69.76717                                                                  | 3.3793         | 0.326       |
| 26. | Octacosane             | 1.030                      | 327.28540                                                                 | 0.3110         | 2.912       |
| 27. | Squalane               | 0.036                      | 460.05000                                                                 | 0.4000         | 7.508       |
| 28. | Dotriacontane          | 2.650                      | 126.50600                                                                 | 1.4601         | 0.820       |
| 29. | Hexatriacontane        | 1.820                      | 256.63290                                                                 | 0.4910         | 1.906       |

Overall average relative deviation values  $p_{\text{pond}}$  is 2.372%

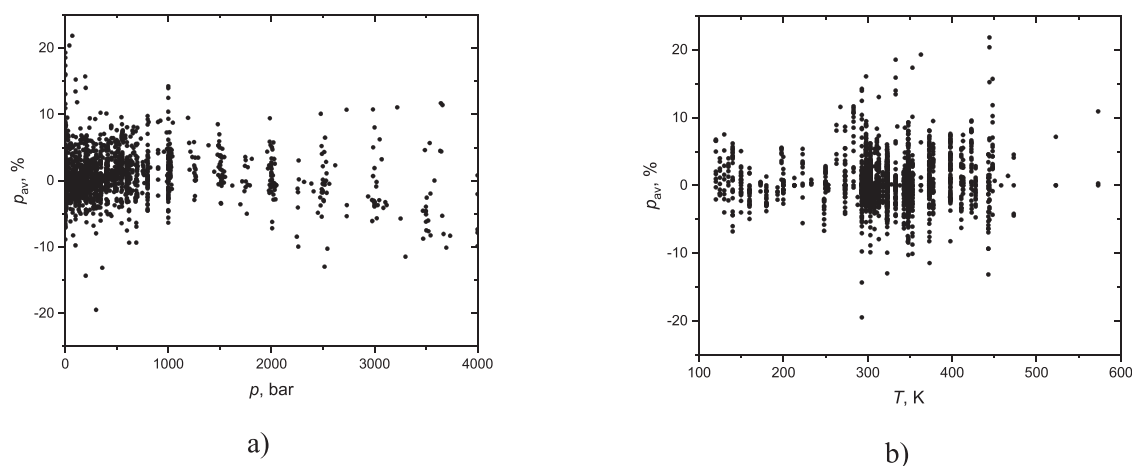


Fig. 2. The absolute average relative deviation values per sets for correlation of liquid hydrocarbon viscosity in function of a) pressure and b) temperature.

Table 5

Prediction results for experimental data not used in the development of the proposed new model, number of compounds,  $l$ , minimum and maximum temperature values,  $T_{\min}$  and  $T_{\max}$ , minimum and maximum pressure values,  $p_{\min}$  and  $p_{\max}$ , and number of experimental data,  $N$ , the absolute average relative deviation values per sets,  $p_{av}$ , and overall average relative deviation values,  $p_{pond}$ .

| $l$                                                            | Compounds                    | $T_{\min}/K$ | $T_{\max}/K$ | $p_{\min}/bar$ | $p_{\max}/bar$ | $N$ | References        | $p_{av}/\%$ |
|----------------------------------------------------------------|------------------------------|--------------|--------------|----------------|----------------|-----|-------------------|-------------|
| 1.                                                             | Pentane                      | 298.15       | 393.15       | 1.013          | 1000.600       | 41  | [159,172]         | 4.890       |
| 2.                                                             | Hexane                       | 298.15       | 298.15       | 1.013          | 706.000        | 16  | [160,161]         | 1.856       |
| 3.                                                             | Octane                       | 298.15       | 473.15       | 1.013          | 2019.000       | 89  | [162,173]         | 1.719       |
| 4.                                                             | Decane                       | 293.15       | 373.15       | 1.013          | 1917.000       | 51  | [162–164,173,174] | 3.388       |
| 5.                                                             | Dodecane                     | 293.15       | 473.15       | 1.013          | 1917.400       | 92  | [162,165–167,175] | 5.554       |
| 6.                                                             | Tridecane                    | 293.15       | 333.15       | 1.013          | 1.013          | 9   | [168]             | 6.884       |
| 7.                                                             | Hexadecane                   | 293.15       | 564.15       | 1.013          | 1028.100       | 141 | [169,170,176–180] | 8.324       |
| 8.                                                             | Eicosane                     | 374.95       | 534.25       | 13.79          | 13.790         | 5   | [171]             | 5.761       |
| 9.                                                             | Isooctane                    | 288.15       | 323.15       | 1.013          | 1.013          | 14  | [181–183]         | 1.373       |
| 10.                                                            | 2,2,4,6,6-pentamethylheptane | 293.15       | 373.15       | 1.000          | 1.000          | 9   | [179]             | 3.955       |
| 11.                                                            | Cyprane                      | 293.15       | 373.15       | 1.013          | 1.013          | 5   | [180]             | 9.398       |
| Overall average relative deviation values $p_{pond}$ is 5.155% |                              |              |              |                |                |     |                   |             |

the prediction results. The correlation results are shown in Table 4 and Fig. 2. No generally conclusion can be made, considering temperature range, pressure range and structure of alkanes. All-thought, maximal errors over 15% are observed at branched alkanes for pressures under 300 bar: squalane (18.533 and 19.479%), pristane (19.294, 17.348, 15.892 and 16.071%), 2,2-dimethylpropane (21.836, 20.371 and 15.2335%).

The absolute average relative deviation and overall average relative deviation values are calculated using Eq. (5–6). The absolute average relative deviation values per sets shown in Table 3 refer to the predicted viscosity results, and in Table 4 refer to the correlated viscosity results.

The model was then tested on experimental viscosity data, which were not used in model development. The prediction results are shown in Table 5.

## 5. Conclusions

In this paper, a new group contribution model based on the Free-Volume Theory for predicting the viscosity values of liquid n-alkanes and i-alkanes was developed. The proposed model allows predicting the liquid dynamic viscosity values with knowledge only of liquid density data at different temperature and pressure, i.e. without knowing any critical values, refractive indices, acentric factor or normal boiling and normal freezing temperature. Experimental values of these quantities are not always available and the use of their predicted values introduces the errors of their prediction into the prediction of dynamic viscosity values. The

model was developed using 2129 experimental data with a temperature range up to 573.15 K and a pressure range up to 4000 bar. Only experimental data from the available literature with experimentally determined density values were used to develop the model, so that errors of correlation or prediction of density values would not affect the development of the new model. Based on the newly developed model, a weighted mean percentage error of 4.897% was obtained for the used experimental data and 5.155% for the 472 experimental data on which the model was tested.

## CRedit authorship contribution statement

**Jovan D. Jovanović:** Formal analysis. **Nikola D. Grozdanić:** Writing – original draft. **Ivona R. Radović:** . **Mirjana Lj. Kijevčanin:** Supervision, Resources.

## Data availability

All data used in this work are literature data and are available for all researchers through the list of references.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgement

This work was supported by the Ministry of Education, Science and Technological Development of the Republic of Serbia (Contract No. 451-03-68/2022-14/200135).

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